Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

Palaash Gang
Independent Researcher
palaashgang@gmail.com

Abstract

A printed claim that the scorecaster breaks online conformal coverage meets its refutation by overlapping authors; this paper joins them and locates the edge of the set that refutation relies on. Conformal PID states long-run coverage for any bounded scorecaster; a corollary keeps it for predictable perturbations of the deployed threshold inside an admissible radius. The claim was acted on; placement and derivation are conceded by name. At the null scorecaster that condition is *tight*. A legal L_1 dead band on the *completed* threshold leaves that set at its radius, and past it miscoverage is 1.000000 against $\alpha = 0.1$. Coverage goes, not only the rate. The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

A refereed paper states that a scorecaster breaks Conformal PID's coverage guarantee. A later one, by the same first and last authors, proves in the same notation that a bounded perturbation of the deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the joining is what this paper adds.

An online conformal method emits a threshold before each observation and moves it. Forecast stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

The record carries a claim and its refutation; this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: "any function of the past" (Theorem 1). Such a readout is already quantified over. The claim reads "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add an arbitrary predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ to the deployed threshold, note that the result "has exactly the CPID error-integration form", and conclude $|E_T| = o(T)$. The same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. Four surfaces were queried on 20 August 2026 for both identifiers and "scorecaster". They are arXiv full text, arXiv abstracts, OpenReview, and the six works Semantic Scholar records citing Li and Rodríguez [2025]. None returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice "arbitrary" and lacking "principled guidance" while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

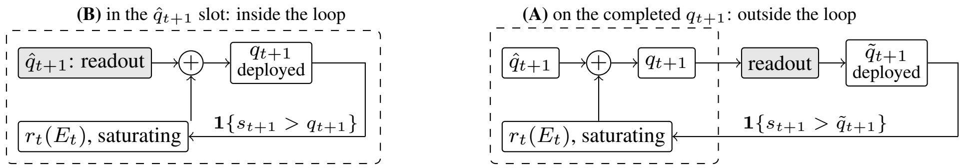

Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator's own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and the deployed sequence leaves that set.

Where that refutation stops, the condition is tight. Corollary 2 is a *retention* result, silent outside its radius, and the question it leaves is whether that radius is sharp. Section 3 exhibits a legal perturbation leaving it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge is the corollary's radius at the null scorecaster. Past it coverage goes, not only the rate: the published sufficient condition is necessary there. Section 4 sets both claims beside four vocabularies for the same movement.

2 Setup

The producer. A forecaster fit before t induces a score s_t, and a threshold q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \le q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1} \sum_{t \le T} \mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale,

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i \le t} (\mathrm{err}_i - \alpha)\Big), \tag{1}$$

their iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \ge b$ with the sign of x once $|x| \ge c\,h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \le t} (\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \le c\,h(T) + 1$, read on realised scores.

One scalar, two readouts, two places. Damping a sequentially updated scalar has two standard readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-\lambda)\hat{q}_{t+1}^{\mathrm{raw}} + \lambda \tilde{q}_t$ [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band, moving only when the target is more than τ away, then pulling back exactly τ [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)** is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits Proposition 2's rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it. What is tight is the radius, not a form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H} |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive distribution $\sum |\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it *jumpiness*, a *convergence index*, *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length. Table 2's caption names ten, so a fresh one is used here.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null. The adversary is specified, not tie-broken. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, and "Sat." the fraction of rounds on which (4) delivers its full $\pm b$. Every number is the *primary* regime's, defined in Appendix A, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. $2{\times}10^5$	miscov. 10^6	$\max_t |E_t|$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	**0.100004**	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	**1.000000**	900,000	60,747	1.0000	0.5

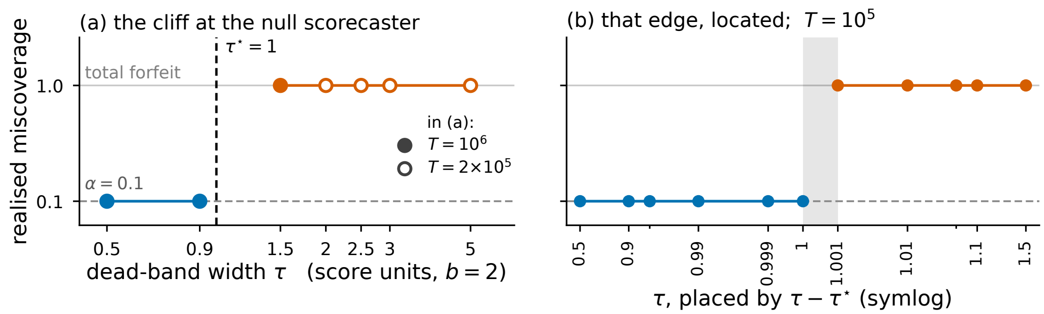

Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b, where $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ is 1. **(a)** The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2{\times}10^5$; blue covers, orange forfeits, and the dashed rule is $\tau^\star$. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^\star$ on a symlog axis whose linear zone is $|\tau - \tau^\star| \leq 10^{-3}$, which magnifies the neighbourhood of $\tau^\star$: visual spacing here is not proportional to true distance in τ. Unlabelled minor ticks mark $\tau = 0.95$ and 1.05. The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings.

3 The edge of the admissible set, measured

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \leq c\,h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight w leaves it; Table 1 prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$. For $w = 0.99$ and 0.999 the excursion is *flat in* T while the bound grows like $\log T$. At $w = 0.999$ the ratio is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. The forfeit's sign is counter-intuitive. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put (4) at risk. It delays a correction already computed, and the accumulator excurses *further*. Smoothing the output does not damp E_t; it feeds it. That 623.70 turns on the adversary's ε, not the counting convention.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so "inside Corollary 2" reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling, and its realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary's best legal play is $b/2$, so coverage is lost when that supremum drops below it, at $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. All 19 dead-band runs agree with that law (Figure 3); only the null scorecaster's edge is located (Figure 2), the others one-sided. **Here** $\sup_x r_t(x) = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^\star = b/2$, one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound. The equally legal $\hat{q} \equiv -b/2$ drops the edge to 0, and there $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the theorem nothing; it moves the edge too (Figure 3) and shrinks the excursion. Under the unbounded tangent integrator Angelopoulos et al. [2023] report using in all their experiments, $\tau^\star$ is unbounded and all three widths there cover. The excursion is 20.10 rather than 623.70, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here.** Outside the set the transition is a step. Past $\tau^\star$ the threshold sits permanently below the reachable score range and $|E_t| = (1-\alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$. The wider bands $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too at $T = 2{\times}10^5$ (Figure 2), so the failing set is $(\tau^\star, \infty)$, open at the left. **The endpoint is where the adversary's specification decides the answer.** At $\tau = \tau^\star$ the deployed supremum equals the adversary's own cap $b/2$, so the clip forces an exact play and the reading settles it: the primary regime covers. Appendix A separates the two readings and prints both. **The failing arms have** $\mathrm{Sat.} = 1.0000$**.** Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop *is* the integrator's output.

What that measures is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the measured $\tau^\star$ is exactly Corollary 2's radius, and crossing it costs coverage rather than the rate. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only: at $\hat{q} \equiv +b/2$ Corollary 2 permits radius 0 while every width run there covers. That asymmetry makes "tight" a located claim.

The constructive reading, and it is the corollary's rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What the measurement adds is that the band has an edge, computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is not a graceful degradation.

4 Where this sits

Four vocabularies name one quantity many ways, and the object that settles its validity (the admissible set around the deployed threshold) sits in the first of them. Table 2 in Appendix B names them and what each contributes. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \le T} q_t \le O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived.

5 Limitations

The evidence is one construction, one adversary, one saturator; and $0.63/(1-w)$ is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(c\,h(T)+1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. One seed, and only the null scorecaster's edge is bracketed**;

the others are one-sided. **What would overturn the claim** is a legal $(r_t, \hat{q})$ pair with $\tau \leq \tau^\star$ that loses coverage, or $\tau > \tau^\star$ that keeps it; the sweep has neither.

References

Lachlan L. H. Andrew, Siddharth Barman, Katrina Ligett, Minghong Lin, Adam Meyerson, Alan Roytman, and Adam Wierman. A tale of two metrics: Simultaneous bounds on competitiveness and regret. In *Proceedings of the 26th Conference on Learning Theory (COLT 2013)*, pages 741–763, 2013.

Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. Conformal PID control for time series prediction. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. doi: 10.52202/075280-1000.

Allen Emmanuel Binny and Anushri Dixit. Who moved my distribution? Conformal prediction for interactive multi-agent systems. arXiv:2511.11567v1, 2025.

Jiadong Chen, Yang Luo, Xiuqi Huang, Fuxin Jiang, Yangguang Shi, Tieying Zhang, and Xiaofeng Gao. IPOC: An adaptive interval prediction model based on online chasing and conformal inference for large-scale systems. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pages 202–212, 2023. doi: 10.1145/3580305.3599396.

George M. Constantinides. Capital market equilibrium with transaction costs. *Journal of Political Economy*, 94(4):842–862, 1986. doi: 10.1086/261410.

M. H. A. Davis and A. R. Norman. Portfolio selection with transaction costs. *Mathematics of Operations Research*, 15(4):676–713, 1990. doi: 10.1287/moor.15.4.676.

Ricarda Duerst, Jonas Schöley, Julia Hellstrand, and Mikko Myrskylä. Calibrating probabilistic forecast paths on past forecast errors: an application to the Finnish total fertility rate. Technical Report WP-2024-016, Max Planck Institute for Demographic Research, 7 2024.

Théo Dupuy, Binbin Xu, Stéphane Perrey, Jacky Montmain, and Abdelhak Imoussaten. Relevance-aware thresholding in online conformal prediction for time series. In *Communications in Computer and Information Science*, pages 196–217, 2026. doi: 10.1007/978-3-032-16708-8_17.

Uwe Ehret. Convergence index: a new performance measure for the temporal stability of operational rainfall forecasts. *Meteorologische Zeitschrift*, 19(5):441–451, 2010. doi: 10.1127/0941-2948/2010/0480.

Sergio Galeani, Sophie Tarbouriech, Matthew Turner, and Luca Zaccarian. A tutorial on modern anti-windup design. *European Journal of Control*, 15(3-4):418–440, 2009. doi: 10.3166/ejc.15.418-440.

Ben Gao, Jordan Patracone, Stéphane Chrétien, and Olivier Alata. Conformal online learning of deep Koopman linear embeddings. arXiv:2511.12760v2, 2025.

Nicolae Gârleanu and Lasse Heje Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013. doi: 10.1111/jofi.12080.

Evgenii Genov, Julian Ruddick, Christoph Bergmeir, Majid Vafaeipour, Thierry Coosemans, Salvador García, and Maarten Messagie. Balancing forecast accuracy and switching costs in online optimization of energy management systems. *Expert Systems with Applications*, 298:129305, 2026. doi: 10.1016/j.eswa.2025.129305.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 69(2): 243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

Rakshitha Godahewa, Christoph Bergmeir, Zeynep Erkin Baz, Chengjun Zhu, Zhangdi Song, Salvador García, and Dario Benavides. On forecast stability. *International Journal of Forecasting*, 41(4): 1539–1558, 2025. doi: 10.1016/j.ijforecast.2025.01.006.

Dongjian Hu, Junxi Wu, Shu-Tao Xia, and Changliang Zou. Distribution-informed online conformal prediction. arXiv:2512.07770v2, 2026.

Ankit Lade, Sai Krishna J., and Indar Kumar. Bias-corrected adaptive conformal inference for multi-horizon time series forecasting. arXiv:2604.13253v1, 2026.

Ruipu Li and Alexander Rodríguez. Neural conformal control for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 18439–18447, 2025. doi: 10.1609/aaai.v39i17.34029.

Ruipu Li, Daniel Menacho, and Alexander Rodríguez. Optimization-based online conformal prediction for multi-step forecasting. arXiv:2508.13362v3, 2025.

Yizhou Min, Yizhou Lu, Lanqi Li, Zhen Zhang, and Jiaye Teng. Questioning the coverage-length metric in conformal prediction: When shorter intervals are not better. arXiv:2601.21455v2, 2026.

F. Pappenberger, H. L. Cloke, A. Persson, and D. Demeritt. HESS Opinions "on forecast (in)consistency in a hydro-meteorological chain: curse or blessing?". *Hydrology and Earth System Sciences*, 15(7):2391–2400, 2011. doi: 10.5194/hess-15-2391-2011.

P. Pinson and R. Girard. Evaluating the quality of scenarios of short-term wind power generation. *Applied Energy*, 96:12–20, 2012. doi: 10.1016/j.apenergy.2011.11.004.

Kandrika Pritularga and Nikolaos Kourentzes. Forecast congruence: A quantity to align forecasts and inventory decisions. SSRN working paper 4711817, 2024.

Pouria Sarhadi. On the standard performance criteria for applied control design: PID, MPC or machine learning controller? arXiv:2503.14379v3, 2025.

Huseyin Tunc, Onur A. Kilic, S. Armagan Tarim, and Burak Eksioglu. A simple approach for assessing the cost of system nervousness. *International Journal of Production Economics*, 141(2): 619–625, 2013. doi: 10.1016/j.ijpe.2012.09.022.

Jente Van Belle, Ruben Crevits, and Wouter Verbeke. Improving forecast stability using deep learning. *International Journal of Forecasting*, 39(3):1333–1350, 2023. doi: 10.1016/j.ijforecast.2022.06.007.

Jente Van Belle, Honglin Wen, Wouter Verbeke, and Pierre Pinson. Stabilizing distribution-free probabilistic forecasts. arXiv:2605.28531v1, 2026.

Rahul Vaze. Simultaneous coverage and efficiency guarantee in online conformal prediction. arXiv:2607.26577v1, 2026.

Xiaoqian Wang and Rob J. Hyndman. Online conformal inference for multi-step time series forecasting. arXiv:2410.13115v2, 2024.

Rochelle P. Worsnop, Michael Scheuerer, Thomas M. Hamill, and Julie K. Lundquist. Generating wind power scenarios for probabilistic ramp event prediction using multivariate statistical post-processing. *Wind Energy Science*, 3(1):371–393, 2018. doi: 10.5194/wes-3-371-2018.

Junxi Wu, Dongjian Hu, Yajie Bao, Shu-Tao Xia, and Changliang Zou. Error-quantified conformal inference for time series. arXiv:2502.00818v2, 2025.

Marco Zanotti. Analyzing the retraining frequency of global forecasting models: towards more stable forecasting systems. arXiv:2506.05776v3, 2026.

Mingyi Zheng, Hongyu Jiang, Yizhou Lu, and Jiaye Teng. Smoothing-based conformal prediction for balancing efficiency and interpretability. arXiv:2509.22529v1, 2025.

Ervin Zsótér, Roberto Buizza, and David Richardson. "Jumpiness" of the ECMWF and Met Office EPS control and ensemble-mean forecasts. *Monthly Weather Review*, 137(11):3823–3836, 2009. doi: 10.1175/2009MWR2960.1.

Code and data

The simulator, its frozen configuration, the `results/` files behind every figure and table, the generators that read them, and this project's audit trail across sessions are at `https://github.com/pa1aash/Turnover-Blind`.

A Supporting measurements

The primary regime. Every number in Table 1 is measured under one deterministic adversary. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$ against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and the distinction is measured rather than argued. Scoring an exact-threshold play *as a miscoverage* reproduces every quantity Table 1 reports: 44 of 44 arm-horizon cells identical, all eleven arms at all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every row, because the adversary's signature move then covers rather than misses. Partial adjustment at $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading. **No row is convention-free with respect to the second**, which is why the adversary is specified rather than tie-broken. At $\tau = \tau^\star$ itself the two readings part: the strict indicator returns 0.000000 where scoring the exact play as a miscoverage returns 1.000000.

Excursions at $T = 10^4$, and where the excursion law holds. Partial adjustment excurses 6.30, 63.00 and 623.70 at $w = 0.9$, 0.99 and 0.999. The last two are flat in T while Proposition 2's bound grows like $\log T$, which is why the ratio falls with the horizon. On the control and the four partial-adjustment arms the excursion fits $\max_t |E_t| \approx \max\{0.5\, h(T) + 0.9,\ 0.63/(1-w)\}$ at every horizon, most loosely at $w = 0.9$, the one arm whose two branches cross inside the horizon range. This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing time constants, the running mean and the dead bands all excurse further than it predicts.

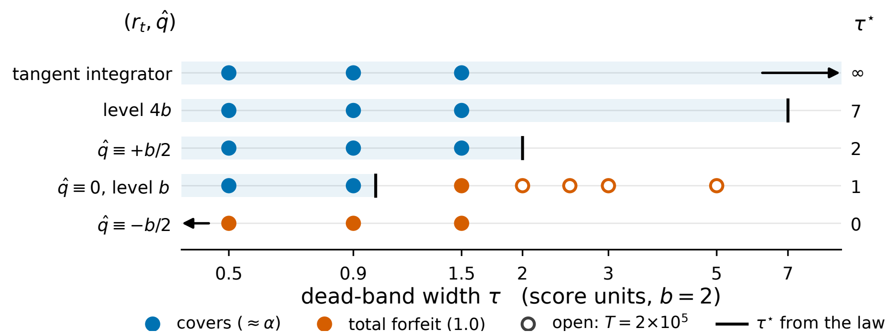

Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by $\tau^\star$, and the shared horizontal axis is τ. Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2{\times}10^5$. The rule on each row is that setting's $\tau^\star = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \le \tau^\star$. Two rules fall off the axis and are drawn as arrows: $\tau^\star = 0$ at $\hat{q} \equiv -b/2$, and $\tau^\star$ unbounded under Angelopoulos et al.'s tangent integrator.

Realised miscoverage at the covering settings of Figure 3. At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.'s tangent integrator the same width returns 0.100001. Raising the saturator's level to $4b$ leaves coverage intact and cuts the $w = 0.999$ excursion from 623.70 to 120.60. That is the first term of $\tau^\star$ moving the edge, not the readout changing behaviour.

B The four vocabularies

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	**Forecast stability**	**Verification**	**Switching-cost online learning**
a distribution-free long-run-coverage guarantee for *any* bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	the scored movement functional, the readout map as post-processing [Godahewa et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2] and [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control *stronger* than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013]